

[← Back](#)**Q1 List Comprehension - Reverse**

10 marks

WAP to display in reverse of numbers from 1 to 10 using List Comprehension
output:
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]

Your Code

```
1 l = [x for x in range(10, 0, -1)]  
2 print(l)
```

Input

stdin input

Output

```
[10, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

[← Back](#)**Q2 List Comprehension - Sum**

10 marks

WAP to find the Sum of First 10 Natural Numbers
output:
55

Your Code

```
1 l = sum([x for x in range(1, 11)])  
2 print(l)
```

Input

stdin input

Output

```
55
```

[← Back](#)**Q3 List Comprehension - Sum of ODD**

10 marks

WAP to find the Sum of Odd Numbers up to 10
output:
25

Your Code

```
1 l = sum([x for x in range(1, 11) if x%2 != 0])  
2 print(l)
```

Input

stdin input

Output

25

[← Back](#)**Q4 List Comprehension - Sum of Squares**

10 marks

WAP to find the Sum of Squares up to 5
output:
55

Your Code

```
1 l = sum([x*x for x in range(1,6)])  
2 print(l)
```

Input

stdin input

Output

55

[← Back](#)**Q5 List Comprehension -Sum of Numbers Divisible by 3**

10 marks

WAP to find the Sum of first 12 Numbers Divisible by 3

Output:

30

Your Code

```
1 l = sum([x for x in range(1,13) if x%3==0])
2 print(l)
```

Input

stdin input

Output

30

[← Back](#)**Q6 List Comprehension - Factorial**

10 marks

WAP to find the factorial of a given number

input:

5

output:

120

Your Code

```
1 import math
2 n = 5
3 l = math.prod([x for x in range(1, n + 1)])
4 print(l)
```

Input

stdin input

Output

120

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Q7 List Comprehension - Sum of Factorials

10 marks

WAP to find the Sum of Factorials

Series: $1! + 2! + 3! + 4!$

Output:

33

Your Code

```
1 import math
2 l = [math.factorial(x) for x in range(1,5)]
3 print(sum(l))
```

Input

stdin input

Output

33

Q8 List Comprehension - Sum of Digits of a Number

10 marks

WAP to find the Sum of Digits of a Number

input:

4567

output:

22

Your Code

```
1 n = 4567
2 l = sum([int(x) for x in str(n)])
3 print(l)
```

Input

stdin input

Output

22

Q9 List Comprehension - Sum of Positive Numbers in a List

10 marks

WAP to find the Sum of Positive Numbers in a List

input:

[10,-5,20,-8,15,-2]

output:

45

Your Code

```
1 n = [10, -5, 20, -8, 15, -2]
2 l=[x for x in n if x>0]
3 print(sum(l))
```

Input

stdin input

Output

45

Q10 List Comprehension - Sum of Squares of Even Numbers

10 marks

WAP to Find the Sum of the Squares of Even Numbers up to 10

(2² + 4² + 6² + 8² + 10²)

OUTPUT:

220

Your Code

```
1 l = [x*x for x in range(1,11) if x%2==0]
2 print(sum(l))
```

Input

stdin input

Output

220